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AEM Tier 2

Nutrient Management Recommendations

Fertilizer Storage

What is the storage duration of fertilizers?

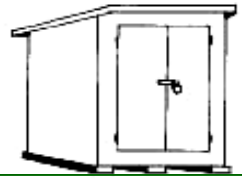
What type of storage is used for dry formulations?

Storage

- Purchase only the amounts of fertilizers required for a single growing season to minimize the necessity for off season storage. Even "seasonal" storage can result in spills or leaks that can potentially pollute groundwater.
- Consult fertilizer labels for any specific storage information.
- Secure the storage of fertilizers away from pesticides.
- Do not store liquid fertilizers in direct sunlight.

Storage Shelving

- Shelving should be located in a dry area away from activities that might damage the fertilizer bags or containers.
- Metal shelves are easier to clean than wood if a spill should occur.
- Store dry fertilizers above liquid fertilizers or store separately to prevent the potential for liquids from above dripping onto the paper container and causing contamination of the product.
- Provide pallets to keep larger bags of fertilizer off the floor.
- Label contents of bulk storage tanks.
- Storage containers in good shape minimize the risk of a spill or leak.



Storage Structure

- A locked storage cabinet or building provides security. Preventing the unauthorized use of fertilizer reduces the chance of accidental spills or theft.
- Dry fertilizer should be stored inside a structure or device capable of preventing contact with precipitation and/or surface water.
- Structure should have an impermeable floor, such as sealed concrete, with a raised circumference curb.
- All liquid fertilizer storage areas should have a secondary containment system that will properly handle and contain spills.
- Secondary containment areas should not have a working floor drain, unless it leads to a collections sump.

References: DSHM-PES-0503, Pesticide Storage Guidelines; NORTH CAROLINA COOPERATIVE EXTENSION SERVIC, document was updated on 9/13/00 by Janet Young. http://osuextra.okstate.edu/pdfs/F-1012web.pdf, http://www.agairupdate.com/aau/aausecurityone.html, http://www.epa.gov/agriculture/ag101/pestfertilizer.html, http://www.ecn.purdue.edu/SafeWater/farmasyst/surveys/factsheets/fertilizer3.htm; http://www.ext.vt.edu/pubs/farmasyst/442-906/442-906.html#L2; http://farmasyst.ifas.ufl.edu/farmasyst/fs3F.pdf

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Nutrient Management Recommendations

What is the condition of the containers?

Storage Containers

- Sound containers are your first defense against a spill or leak. There should be **no** holes, tears, weak seams or leaks in the bags or containers.
- Tanks or bags need to have labels that are clearly visible and readable.
- Common sense suggests keeping fertilizers in their original containers with their attached labels. Food or beverage containers are never used for storage.
- Keep fertilizers dry and out of the way of activities that could rip open a bag or puncture a liquid storage container.
- Treat dry fertilizer impregnated with a pesticide as a pesticide and store with your other pesticides; under cover and protected from the elements.
- Store plastic containers of liquid fertilizers out of direct sunlight.
- Liquid fertilizer tanks should have secondary containment. The level of liquid fertilizer in the container should be able to be determined readily.

Important Points:

Buy only the amount of product you need to apply. Keeping an accurate inventory will help you decide on how much fertilizer you need to purchase.

Avoid long term fertilizer storage.

Past field application records of rate and effectiveness will help you decide what type of product to purchase.

"Reducing fertilizer use and loss makes financial as well as environmental sense."

Reduced fertilizer use & loss equals:

- *Reduced costs*
- *Reduced storage needs*
- *Reduced risk to groundwater*

Container Disposal Practices: **Proper offsite disposal practices are essential to avoid risking contamination that could negatively impact ground and surface waters.**

References: DSHM-PES-0503, Pesticide Storage Guidelines; NORTH CAROLINA COOPERATIVE EXTENSION SERVIC, document was updated on 9/13/00 by Janet Young;Kentucky Cooperative Extension Service, KY.A.Syst publication., http://greenhouses.cals.cornell.edu/BMP2_FertStorAndNutrMngt.html; http://www.ext.vt.edu/pubs/farmasyst/442-906/442-906.html#L6

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Nutrient Management Recommendations

What security measures are taken at the storage area?

Fertilizer Storage Security Measures:

✓ Security should be provided by use of locks, fencing and/or lighting.

✓ Signs or labels outside of the building indicating that the cabinet or building is a fertilizer storage area.

✓ Storage cabinet or building should be locked when not in use to prevent access by unauthorized personnel, children or animals. Access is restricted to trained personnel.

✓ Fertilizers should not be stored in the direct presence of fuel products or pesticides due to the increased potential for explosions as well as disposal problems.

✓Provide local fire-fighters with important information as to lay out of structure and list of all types of fertilizers stored in the building.

✓ Minimize the amount of fertilizer stored and handled at any time.

✓Sound containers are your first defense against spills, leaks or tampering.

✓ Storage areas should be inspected at least annually by the owner or person responsible for the fertilizer to ensure safe storage of fertilizers and to minimize mishaps.

✓ An inventory should be maintained to keep track of fertilizer purchases and their uses.

✓ Keep fertilizers and pesticides in separate containment areas.

✓Keep fertilizers in their original containers with their attached labels. Labels should be clearly visible and readable.

✓ There should be no holes, tears, weak seams or leaks.

✓Bags with tears, holes or leaks should be repackaged, with label clearly visible.

✓ Level of fertilizer liquid in the containers should be able to be determined readily.

✓ Treat dry fertilizer impregnated with a pesticide <u>as a pesticide</u>.

References: DSHM-PES-05-03 Pesticide Storage Guidelines; NYS Department of Environmental Conservation Pesticide Safety Education Core Manual, http://greenhouses.cals.cornell.edu/BMP2_FertStorAndNutrMngt.html

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Nutrient Management Recommendations

What is the distance from the fertilizer storage to the nearest surface waterbody or well?

What is the distance from the mixing/loading area to wells, surface water and water courses?

Storage Distance from Water

★Storage site should be selected that minimizes potential for contamination of surface or groundwater by drainage, runoff, or leaching.

★Fertilizer storage areas should be located a minimum of 100 feet from any private water well and surface water bodies and 500 feet from wells used for public water supply. (NRCS and NYS Public Health Standards, 2005)

★Locating the storage area down slope of a well or surface waterbody will decrease the risk of possible contamination.

★Storage areas should be inspected at least annually by the owner or person responsible for the fertilizer to ensure safe storage of fertilizers and to minimize mishaps.

Mixing & Loading Area Distance from Water

★Avoid mixing and loading fertilizers within 100 feet from any private water well and surface water bodies and 500 feet from wells used for public water supply. (NRCS and NYS Public Health Standards, 2005)

★Locating the mixing & loading area down slope of a well or surface waterbody will decrease the risk of possible contamination.

★Avoid mixing and loading on gravel driveways or other surfaces that allow spills to soak quickly into the soil.

★Use a nurse tank to transport water to the application site.

References: NRCS AMF standard NY-702 requires a minimum of 100 feet from private wells or surface water bodies and 500 feet from wells used for public water supply. Natural Resources Conservation Service Conservation Practice Standard 2004; Core Manual Pesticide Safety Education, http://www.omafra.gov.on.ca/english/environment/efp/infosheet_4.htm#41, http://www.epa.gov/agriculture/ag101/pestfertilizer.html

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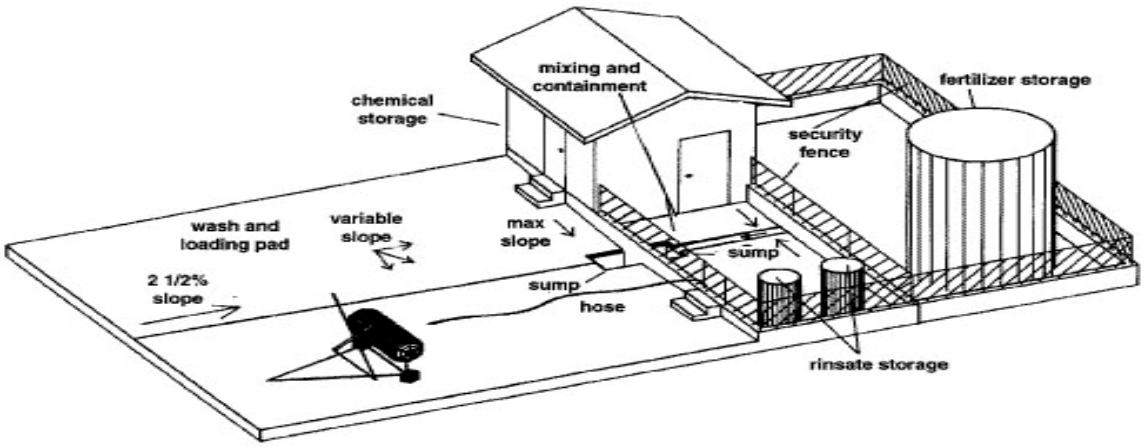
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Mixing & Loading Area

- Ideally, the mixing and loading of fertilizers as well as the rinsing of equipment should be done on an impervious surface (containment pad) with curbs. Be sure that any run-off water from the pad does not flow into ditches, streams, well heads or any nearby waterbodies.
- A concrete mixing and loading pad with curbs keeps any spills contained, allowing for a quick and easy cleanup of the spilled fertilizer and results in minimal contamination of the groundwater. The size of the pad depends on the type of equipment you use. It should provide space around the parked equipment for washing and rinsing.
- A containment sump allows for the collection of a spill and then transfer to storage.
- If you don't have a containment pad, vary the location of mixing and loading. **Clean up spills** (even small ones) **immediately**. **Do not let the material soak into the ground.**
- The mixing pad should be large enough to easily accommodate your equipment and contain spills 110 percent of the volume of the largest container to be stored or loaded in the area. Locate the pad next to the storage area. At sites where runoff water could reach a well, construct a diversion so runoff is directed to another area.



References: DSHM-PES-05 Pesticide Storage Guidelines, NY State DEC Program Policy; Cornell University PMEP; Reducing the Risk of Groundwater Contamination by Improving Pesticide Storage and Handling; NORTH CAROLINA COOPERATIVE EXTENSION SERVICE This document was updated on 9/13/00 by Janet Young, http://muextension.missouri.edu/explore/envqual/wq0677.htm; http://www.purdue.edu/dp/envirosoft/fert/src/improvm.htm

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Nutrient Management Recommendations

Nitrogen Management Practices

How are N fertilizer rates determined?

Are N credits considered?

N Fertilizer Rates

- It is necessary to determine, as accurately as possible, the quantity of nitrogen necessary for maximum economical crop production. Apply only this quantity of nitrogen to prevent over-fertilization.
- To determine the quantity of nitrogen that should be furnished by commercial fertilizer, all the nitrogen sources must be considered. Consider all sources of nitrogen (soil organic matter, legumes, cover crops). Do not use fertilizer from these sources as insurance fertilizer. Unnecessary fertilization is too expensive from the standpoint of both profits and the environment.
- The guidelines of fertilizer rates can be found in the Cornell Guides in either print or web forms (see below for additional information). If the rates are applied as recommended, the losses of fertilizer nutrients into the environment are relatively small and optimum economical production is achieved.

Guidelines for Efficient Use of Nitrogen

- Use cover crops to retain nitrogen and other nutrients. They also limit leaching.
- Account for nitrogen from organic matter, cover crops, compost, and manure when planning on fertilizing a field.
- Limit nitrogen applications prior to planting, and avoid deep plow-downs.
- Consider using the pre-sidedress soil nitrate test (PSNT) to determine nitrogen contributions from non-fertilizer sources.
- Apply nitrogen just prior to the plant's peak needs.
- To apply nitrogen most efficiently band either at planting or as a sidedress.
- Use plastic mulch to limit leaching .
- Avoid "insurance" applications of nitrogen.
- Avoid over fertilization which will lead to leaching.
- Maintain proper pH.

Online Sources for Determining N Fertilizer Rates include:

2007 Cornell Guide for Integrated Field Crop Management
http://ipmguidelines.org/fieldcrops/

2006 Integrated Crop & Pest Management Guidelines, for Commercial Vegetable Production
http://www.nysaes.cornell.edu/recommends/

2006 Cornell Guide Integrated Management of Greenhouse Floral Crops
http://www.greenhouse.cornell.edu/pestsdis/pdfs/2006guidelines/01front.pdf

2006 Cornell Pest Management Guidelines for Commercial Tree Fruit Production
http://www.nysaes.cornell.edu/ent/treefruit/

Print Copies can be purchased at: http://store.cce.cornell.edu/

Or call your local extension educator for help or more information

References: 2006 Cornell Guide for Integrated Field Crop Management; http://nmsp.css.cornell.edu/publications/articles/extension/

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Nutrient Management Recommendations

Nitrogen Management Practices

Are soil and/or tissue test results used?

Soil Tests

Soil Tests :

- Measure the "plant available" nutrients in the soil.
- Measure the soil pH.
- Measure the percent of organic matter in the soil.
- Samples should be taken, at a minimum, every two to three years.

Tissue Tests

Tissue Tests:

- Alone, tissue tests can be used for making fertilizer recommendations for certain crops. Or plant tissue analysis can be used in combination with soil test information. The combined method is the recommended approach for diagnosing nutrient deficiencies and determining fertilizer requirements.
- Analysis determines the present levels of elements in plant material. The elements analysis may include nitrogen, phosphorus, potassium, magnesium and calcium.

Soil Testing Labs

Cornell University
Cornell Nutrient Analysis Laboratory
804 Bradfield Hall, Ithaca, NY 14853
Phone: 607-255-4540 Fax: 607-255-7656

Benefit of Soil & Tissue Testing

Grower Benefits:

- Ensures fertile soil without excess fertilizer application.
- Reduces costs from over-fertilization.
- Helps produce optimal economic yields.
- May aid in identifying deficiencies with in your crop.
- Takes the guesswork out of fertilizer recommendations.
- Provides guidelines for amount of lime and fertilizer needed and the timing, method, and placement needed to produce an optimal crop.
- Allows you to follow changes in your soil's properties over time.

Environmental Benefit:

- Minimizes nutrient losses.
- Ensures fertile soil without endangering the environment.
- Reduces water pollution because of over-fertilization and improper timing or placement of fertilizer.

Contact your local CCE educator for additional reputable labs with knowledge about local soil conditions.

References: Cornell Cooperative Extension The Integrated Crop and Pest Management Guidelines for Vegetables 2004
2006 Cornell Guide for Integrated Field Crop Management, http://www.aasl.psu.edu/plant_tissue_prog.html

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Nutrient Management Recommendations

Nitrogen Management Practices

Are fertilizer applicators calibrated?

Spreader Calibration

The two main items to consider when calibrating a spreader are:

1) Distribution pattern of the spreader

2) Product application rate.

The distribution pattern is the pattern the product makes as it settles on the ground after it is thrown out by the spreader's impeller. The application rate will differ for each product.

Helpful reminders when calibrating fertilizer spreaders:

1) **Check to make sure all parts are working properly.**
2) **Calibrate the spreader using the material you will be applying. Fill the hopper only to the level you will have when the material will actually be applied.**
3) **Calibrate the spreader over a tarp or over a sweepable surface so that the fertilizer can be cleaned up and reused.**

Some basic tips for maintaining a spreader:

1) **Check the spreader to make certain all the parts are functioning properly.**
2) **Empty the spreader after each use.**
3) **Wash the spreader and allow to dry. Keep the impeller clean.**
4) **Operate the spreader at a constant speed.**
5) **Lubricate the spreader at recommended intervals.**
6) **Store the spreader in a cool, dry place with no material in it.**

"Over- application can not only be expensive and wasteful, but could be damaging our water resources."

References: NORTH CAROLINA COOPERATIVE EXTENSION SERVICE, *This document was updated on 9/13/00 by Janet Young, http://www.uog.edu/CALS/site/users/turf/calibrate.html, http://turf.arizona.edu/tips197.html*

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Calibration
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Calibrating

Preparing to Calibrate:

Granular application is usually done with the pesticide or fertilizer that will be applied. Extra care needs to be taken when handling any pesticide product; remember to minimize worker exposure and take precautions against spills. To prepare for calibration follow these steps:

1. Before you begin carefully inspect the equipment to ensure all parts are in proper working order.
2. Determine what type of application is required for the product:
   - Broadcast: treats all area (includes band applications based on broadcast rates).
   - Band: treats only the area under the band.
   - Row: treats along the length of the row.
3. Determine the application rate needed:
   - Broadcast: pounds per acre
   - Band: pounds per acre treated with band width.
   - Row: pounds per acre or pounds per 1,000 feet of row length.
4. What type of drive system does the applicator use?
   - Independent: used PTO, hydraulic, or electric motor drive.
   - Ground Drive: uses ground driven wheel.
5. Regardless of how the application rate is expressed or type of application, calibration is easier if the rate is expressed in terms of pounds per foot or row length. Use one of the following equations to determine the correct row rate in pounds per foot.
6. Choose a calibration distance to work with and measure a test course of this distance in the field you will be working in, choose an area that is representative of the overall field conditions. The calibration distance should be at least 50 feet but no more than 500 feet.; the longer the distance the more accurate the calibration.

For broadcast and row applications (Application Rate = lb/ac):	*For banded applications (Application Rate = lb/ac of Band Width):*	*For directed (row) applications (Application Rate = lb per 1,000 ft):*
Row Rate, lb/ft =	**Row Rate, lb/ft =**	**Row Rate, lb/ft =**
$\frac{\text{Application Rate x Row Width (ft)}}{43{,}560}$	$\frac{\text{Application Rate x Band Width (ft)}}{43{,}560}$	$\frac{\text{Application Rate}}{1{,}000}$

7. Calculate the weight of material that should be collected for the calibration distance.
   Use the formula: **Weight Collected = Row Rate x Calibration Distance**
8. Select a ground speed appropriate for the crop and equipment being used. Maintaining an accurate and consistent speed is very important, consult your equipment manual for a recommended speed.
9. Set your equipment according to the recommendations from the equipment or chemical manufacturer. Most manufacturers provide rate charts to determine the correct orifice setting or rotor speed for each applicator. Fill the hopper at least half full to represent the average capacity for calibration.
10. Attach a collection container to each outlet on the applicator; you should be able to collect all the material discharged from the applicator. Locate a scale capable of weighing the samples collected in calibration; remember an accurate scale is important.

References: Sanders, Dr. D.C., *2007 Southeastern Vegetable Crop Handbook*,

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Calibrating A Granular Applicator

Calibration Methods:

Distance Method

The distance method is preferred by most applicators because it applies to any type of granular machine and is rather easy to perform.

1. On the test course selected in the field, collect the output from the applicator in a container as you travel the course, weight the collected material. Record the time required to travel the course. Run this course twice, once in each direction. Average the results for the material collected and time from the two run together.
2. Determine the weight of the product that should be collected for the calibration distance.
   Weight Collected (lb.) = Row Rate (lb/ft) x Calibration Distance (ft.)
3. Compare the weight of the product collected to the weight expected for the calibration distance. If the rate differs by more than 10% adjust the orifice, rotor speed, or ground speed and then repeat the test. Repeat the procedure until the error is less than 10%

Time Method

The time method is similar to sprayer calibration and can be used for applicators driven by PTO, hydraulic, or electric motors.

1. On the test course selected in the field, record the time required to travel the course, do not have the spreader on. Run the course twice, once in each direction, and average those results. Consistent and accurate ground speed is very important for good calibration when using the time method.

2. With the equipment parked on a sweepable surface or have a collection bucket underneath, set the orifice control as recommended and run the applicator for the average time it took to run the distance of the field. Weigh the output of the applicator for the time measurement.

3. Determine the weight of the product that should be collected for the calibration distance.
   Weight Collected (lb) = Row Rate (lb/ft) + Calibration Distance (ft)

4. Compare the weight of the product actually collected during the time it took to cover the calibration distance to the weight expected for the calibration distance. If the rates differ by more than 10%, adjust the orifice, rotor speed, or ground speed and repeat. Bear in mind, speed adjustments are not effective for ground-driven equipment. Repeat the procedure until the error is less than 10%.

References: Sanders, Dr. D.C., *2007 Southeastern Vegetable Crop Handbook*,

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Calibrating a Broadcast Spreader

Broadcast spreaders include machines designed to apply materials broadcasted across the surface of a field. They include drop, spinner, and pendulum spreading devices.

Calibration Methods

There are two common methods used to calibrate broadcast spreaders they are the discharge and pan methods.. The first method, the discharge method requires collecting and measuring the total discharge from the spreader as it runs across a test course. The pan method is used on centrifugal and pendulum spreaders. It determines the pattern shape and swath of the spreader which are then used to determine the application rate.

Discharge Method

1. Determine the test distance to use. The longer distances may give better accuracy but may be difficult to manage, a distance of 300 to 400 feet is adequate. Use shorter distances if necessary to avoid collecting more material than you can handle or weigh.
2. Set the ground speed; be sure to maintain a constant ground speed at all times.
3. If using a **ground drive spreader**, attach a collection bin to the discharge chute or under the outlets and collect all the material discharged from the spreader as it runs across the pre-determined test distance. If using an **independent drive spreader**, record the time required to run the test course. Park the spreader at a convenient location (preferably one with a sweepable floor) and measure the discharge from the spreader for the time measured on the test distance. The course should be run twice and the times averaged for better accuracy.
4. Calculate the application rate (pounds per acre):

$$\textbf{Application Rate, lb/ac} = \frac{\text{Weight Collected (lb)} \times 43{,}560}{\text{Distance (ft)} \times \text{Swath (ft)}}$$

5. Compare the application rate measured to the rate required. Adjust and repeat as necessary.

Pan Method

1. Place pans in the field across the swath to be spread. Pans should be evenly spaced to cover the full swath. One pan should be at the center of the swath with equal numbers of pans on each side. Use enough pans to get a good measurement, eleven is sufficient more is better.
2. Make three passes with the spreader using the driving pattern to be used in the field. One pass should be directly over the center pan and the other passes at the recommended distance and lane spacing to the right and left of the center pass.
3. Combine the material collected in the pans and determine the weight or volume collected. Divide the weight or volume by the number of pans used to determine the average weight or volume per pan.
4. Calculate the application rate.

If you are measuring the weight in the pans in grams:

$$\textbf{Application Rate, lb/ac} = \frac{13{,}829 \times \text{Weight (grams)}}{\text{Pan Area (inches2)}}$$

If you are measuring the volume in the pans in cubic centimeters (cc):

$$\textbf{Application Rate, lb/ac} = \frac{13{,}829 \times \text{Bulk Density (lb/ft3)} \times \text{Volume (cc)}}{\text{Pan Area (inches2)} \times 62.4}$$

5. Compare the rate measured to the rate required, make adjustments and repeat if necessary.

References: Sanders, Dr. D.C., *2007 Southeastern Vegetable Crop Handbook*,

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Corn Planter Fertilizer Calibration

Collect fertilizer in a container (a calf bucket works nicely) or bag tied to the fertilizer drop tube for one row while traveling 1/50 of an acre. (The distance to travel is given in the chart below.)

Weigh the amount of material collected and multiply by 50.

Check against the desired application rate. Make final adjustments and then check again.

Corn Planter Seed Calibration

Fill the seed hoppers at least half full to simulate average planter weight.

Mark the distance equal to 1/1000 acre. (Use the chart below to determine the appropriate distance.)

Plant the measured distance at the speed you intend to use during planting. To simulate accurate field conditions, start planting before reaching first mark and continue past last mark.

Multiply the number of seeds in one row by 1000. This equals the planting population for one acre.

To be accurate, check each row.

Check average distance between seeds to determine accurate seed placement. (If planting 34,000 seeds/ac, there should be 6" between each seed.)

Length of Row to Equal 1/50 and 1/1000 Acre

Row Spacing	1/50 Acre	1/1000 Acre
28"	373'	18' 8"
30"	349'	17' 5"
32"	327'	16' 4"
34"	308'	15' 4"
36"	290'	14' 6"
38"	275'	13' 9"
40"	261'	13' 2"

References: Beckman, Jen, Getting Ready for Corn Planting. Lewis County Ag Digest, April 2006, page 13.

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Nutrient Management Recommendations

Nitrogen Management

Are vegetative filter areas used?

Vegetative Filter Areas

Definition:
A strip of area of vegetation for removing sediment, organic matter, and other pollutants from runoff and wastewater.

Purpose:
To remove sediment and other pollutants from runoff or wastewater by filtration, deposition, infiltration, absorption, adsorption, decomposition, and volatilization, thereby reducing the pollution and protecting the environment.

Conditions Where Filter Strip Practices Apply:

The filter strip practice applies (1) on cropland at the lower edge of fields or on fields adjacent to streams, ponds, and lakes or above conservation practices such as terrace or diversions, (2) in areas requiring filter strips as part of a waste management system to treat polluted runoff wastewater, (3) on forest land where filter strips are needed as part of a forestry operation to reduce delivery of sediment into waterways.

Some of The NRCS Standards for Vegetative Filter Areas/Strip Include:

- The filter area/strip area should be at least 35 feet wide.
- The filter area/strip should have a level cross-section that insures water flow over the area.
- The area/strip should be placed so that the lower edge of the filter area is a minimum of 25 feet away from the surface waterbody.
- The filter strip can be placed where vigorous dense vegetation exists or can be established.
- The filter strip should have a finished grade no less than 2% and no more than 12%.
- Clippings from a filter area need to be removed from that filter area.

"Well maintained grassy buffer areas have been shown to significantly reduce the runoff of residues from mixing & loading locations and from treated areas in the field".

References: NRCS-USDA; http://www.nrcs.usda.gov/feature/outlook/Soil%20Quality.pdf; http://www.nrcs.usda.gov/feature/buffers/BufrsPub.html#CRPbuff_3Anchor; Core Manual Pesticide Safety Education, References: USDA Natural Resources conservation Service, Conservation Practice Standards for New York; Bayer Crop Science, Best Management Practices for Long Island, New York , NRCS Conservation Practice Standard-Filter Strip (code 393)

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Nutrient Management Recommendations

Nitrogen Management

How is crop residue managed?

Managing Crop Residue

" Runoff and erosion from agriculture lands is recognized as a serious nonpoint source pollution problem on Long Island"

Agronomic measures that increase soil surface protection and decrease soil disturbance, are the preferred ways of addressing erosion concerns.

These include:

Reduced Tillage

No/zone Tillage

Cover Cropping

Rotations with Perennial Crops

Adding organic matter such as compost:

Improves soil aggregation

Increases infiltration

Reduces runoff and erosion

Ways to increase soil protection and decrease soil disturbance:

- ✓ Cover Cropping
- ✓ Reduced Tillage
- ✓ No/Zone-Tillage
- ✓ Crop Rotation
- ✓ Adding Organic Matter:
  - …….. compost
  - …….. manure
  - …….. cover crop

Chisel Tillage is a practice that increases infiltration and reduces runoff and erosion.

Benefits of Chisel Tillage Include:

- Lower costs
- Less power and energy requirements than moldboard plows (10-15 hp per shank)
- Increased flexibility in depth of tillage (5-12 inches and deeper with special shanks)
- Increase in operating speed and depth equals less residue cover
- Disks mounted on front frame—chop residues and prevent build up and clogging of shanks

References: 2004 Cornell Guide for Integrated Field Crop Management; 2006 Cornell Guide for Integrated Field Crop Management

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Nutrient Management Recommendations

Part 2:
Other Nutrient Management Concerns

Are fertilizer application records maintained?

Do you keep complete records of fertilizer applications?

Good Record Keeping Demonstrates Good Management Practices

Benefit of Keeping Complete Fertilizer Records:

- **Improve Efficiency:** Keeping records on nutrients applied to crops by individual field allows operations to know what has been done and how to plan for the future.
- **Environmental Impact:** As growers apply and use production inputs more efficiently, less material will leave the field through surface and groundwaters. Reduced environmental impact can only improve the public's perception and acceptance of producers' agricultural practices.
- **Levels of Production:** By recording & analyzing the amount of fertilizers applied to crops & the resulting crop yields, growers can determine the effectiveness of the fertilizer that was applied.
- **Keep records of nutrient sources, application dates, rates, and methods:** Collected data helps create accurate management plans.

Fertilizer Records Should Contain:

- ☐ Amount of fertilizer purchased along with the date of purchase and the supplier.
- ☐ Date(s) of nutrient application(s).
- ☐ The crop acreage receiving a nutrient application. The rate per acre.
- ☐ The nutrient composition of fertilizer or other nutrient-supplying material used. (Cover crop residues grown in the field and left to recycle nutrients are considered as nutrient additions.)
- ☐ Amount of nutrient-supplying material applied per acre.
- ☐ Method of application and placement of applied nutrients. (i.e., broadcast and incorporated, broadcast and not incorporated, subsurface-banded, surface-banded, soil injected, applied through an irrigation system, etc.)
- ☐ Previous crop grown and yield harvested.
- ☐ Most recent soil fertility test (s) and/or plant tissue analysis reports.

References: Cornell Cooperative Extension Agronomy Fact sheet Series, Fact Sheet 2;University of Kentucky, Guidelines for Adopting Precision Agricultural Practices; University of Maryland College of Agriculture & Natural Resources and Maryland Department of Agriculture; University of Tennessee, Agricultural Extension Service

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Nutrient Management Recommendations

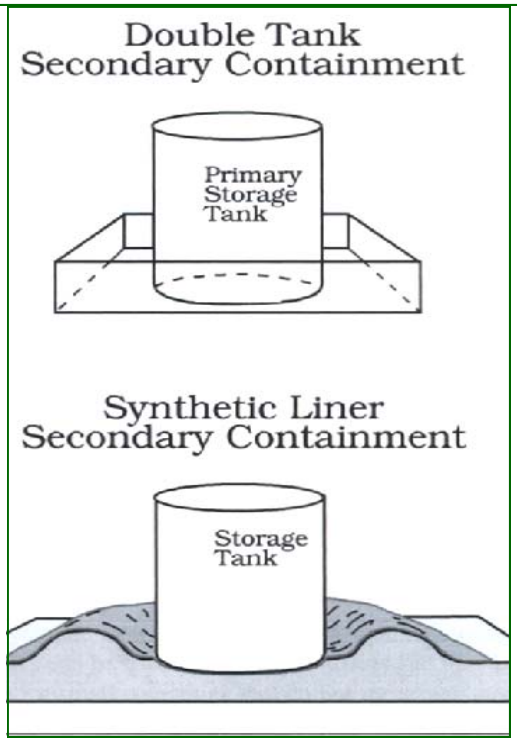
Liquid Fertilizer Storage

What type of storage is used for liquid formulations?

Guidelines for Liquid Fertilizer Containment

Secondary Containment:

A back-up system built around pesticide and fertilizer containers to capture products that may leak or spill.



- Liquid fertilizer should be stored in containers approved for and compatible with the fertilizer being stored.
- All liquid fertilizer storage areas should have secondary containment that will properly handle and contain spills. Containment area floor should be constructed to prevent the absorption or loss of fertilizer.
- Secondary containment areas should not have a working floor drain unless it leads to a collection sump.
- All liquid fertilizer storage containers should be labeled properly.
- Containers, valves, gauges and piping should be made of materials compatible with the products being stored. Backflow protection is recommended when liquid fertilizers are associated with any water supply.
- The level of the liquid in the containers should be able to be determined readily.

Benefit of Secondary Containment:

Protects the environment from accidental leaks and spills of bulk liquid storage tanks by preventing spills from entering the soil and possibly ground or surface water.

References: Colorado State University Cooperative Extension, 1994, Bulletin #XCM-178

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AEM Tier 2
Nutrient Management
Recommendations

Liquid Fertilizer
Mixing & Loading Practices

Where is the water source?

Recommendations Mixing & Loading Liquid Fertilizer

The greatest chance for groundwater contamination problems are associated with mixing and loading of fertilizers near wellheads or surface water. Avoid mixing and loading agrichemicals near your well or any public drinking water supply wells. Fertilizers applied in close proximity to wells can contaminate the ground-water by moving through the wells directly into the ground water supply.

Protect Water Sources by…

- **using a separate water tank such as a nurse tank**, to transport water to the mixing and loading area. This is the best way to prevent direct contamination of a well head. (Increasing the water hose length so the spray unit can be filled farther from the water source is another option to consider.)
- **making sure the water hydrant** is at least 100 feet from private wells and surface water-bodies, and 500 feet from public water supply wells. The fertilizer mixing and loading areas also need to be placed at least 100 feet from private wells and surface water bodies, and 500 feet from wells used for public water supply*.

These standards apply to areas where fertilizers are mixed, loaded and where equipment is cleaned. The mixing and loading area should also be located outside the 100 year flood plain, wetland areas, and downslope from any well, surface water and/or watercourse.

Key Points to remember about Mixing & Loading Practices

- Avoid mixing and loading fertilizers near your well or any public drinking water supply wells. One way to do this is to use a nurse tank to transport water to the mixing and loading site. The site should be a distance of 100 feet downslope of a well or surface waterbody is suggested*. Ideally the mixing site should be rotated within the field of application to avoid build-up of fertilizers in the soil.
- Avoid mixing and loading on gravel driveways or other surfaces that allow spills to sink quickly through the soil.
- Always have someone watching the fill tank and the lines when transferring liquid fertilizers.

*According to NRCS Agrichemical Mixing Facility standard NY-702

References: Natural Resources Conservation Service Conservation Practice Standard 2004; Core Manual Pesticide Safety Education; Cornell University PMEP; Cornell Cooperative Extension Suffolk County Best Management Practices for Long Island Growers

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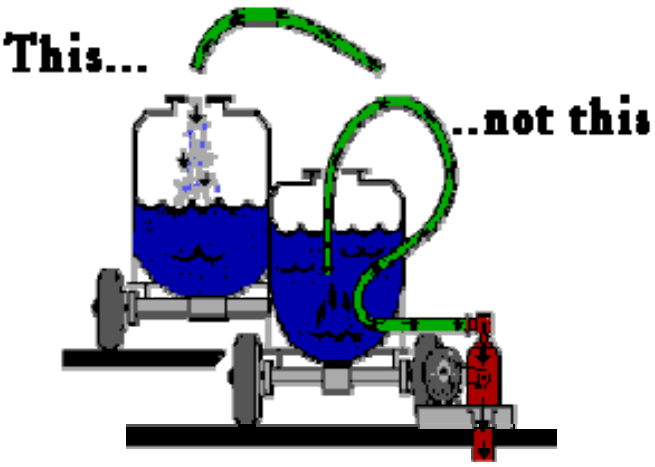
Nutrient Management Recommendations

Liquid Fertilizer
Mixing & Loading Practices

What type of backflow prevention is used during filling?

GUIDELINES FOR FILLING LIQUID FEED TANKS:

- Install a Reduced Pressure Zone (RPZ) devise to avoid contamination.
- **Never** put the hose in the application tank.
- Provide an air gap (twice the diameter of the fill hose) between the hose end and the top of the sprayer tank.
- Use a nurse tank to supply water for mixing so that mixing will occur away from wells and prevent backflow of fertilizers into wells.



Source:
Protecting Our Groundwater—
A Grower's Guide

Air - Gap to Prevent Backflow

Definitions:

Air-Gap: An air-gap separation is an unobstructed physical separation through the free atmosphere between the lowest opening from any pipe or faucet supplying water to a tank, plumbing fixture, or other device and the flood level rim of the receptacle. The differential distance shall be at least double the inside diameter of the supply pipe. In no case shall the air-gap be less than one inch.

Anti-Siphon Device: Prevents contamination of water, which can occur when a loss of pressure in the main water line creates a backflow of contaminated water into the water supply system.

Reduced Pressure Zone Device (RPZ): must contain a minimum of 2 independently-acting check valves, with an automatically-operated pressure differential relief valve located between the 2 check valves. The unit must include tightly closing shutoff valves located at each end of the device. Each device must also be fitted with properly located test locks.

Definitions: **NYS Department of Environmental Conservation**



References: Cornell University PMEP; Cornell Cooperative Extension Suffolk County Best Management Practices for Long Island Growers; Colorado State University Cooperative Extension, 1994. Colorado State University Cooperative Extension, 1994, Bulletin #XCM-178

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Nutrient Management Recommendations

Liquid Fertilizer
Mixing & Loading Practices

What is the condition of the piping, valves and tanks?

Valve Maintenance

Check valves for good operating and proper back flow sealing prior to operating the complete system.

Provide lockable shutoff valves on outlets of all bulk liquid fertilizer tanks for security and to isolate system components for maintenance or emergencies.

Use corrosion resistant valves made from stainless, polypropylene or Kevlar.

Minimize the number of valves to operate the system to reduce cost and potential leaks.

Tank Maintenance

Elevating storage tanks allows tank bottoms to dry off periodically and reduces corrosion.

Provide 3' clearance around tanks so there is access to the entire tank to check for leaks.

Routinely paint metal tanks, steel frames, grates and anchors to reduce corrosion.

Routinely Inspect and Maintain the Following:

- Storage tanks (corrosion, mechanical damage, leaks) ❑
- Tank valves and fittings ❑
- Tank supports and anchors ❑
- Containment area ❑
- Emergency /safety equipment ❑
- Concrete joints and sealants ❑
- Concrete floor and wall surfaces ❑
- Sumps and sump pumps ❑
- Protective coatings and paint ❑
- Electrical systems and controls ❑
- Sight gauges ❑

Plumbing Maintenance

Protect plumbing from possible ice damage due to trapped water in the containment or in pumps.

Drain and rinse out pumps in cold weather.

Elevate pipes and provide a gap between pipes so dirt, leaves, etc. will not be trapped under or around the pipes.

Use detachable hoses instead of hard plumbing to isolate storage tanks from other plumbing.

References: Agricultural Chemicals and Groundwater Protection, Colorado State University Cooperative Extension, 1994

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Nutrient Management Recommendations

Liquid Fertilizer
Mixing & Loading Practices

What type of handling system is used?

How is the applicator cleaned and rinsate disposed of?

Mixing & Loading Handling System

Recommendations:

- Use a closed handling system, where the fertilizer is directly transferred from the storage container to the applicator equipment, such as by a hose. This will ensure that humans and the environment are never inadvertently exposed to the chemical.
- If liquid spillage occurs, recover as much of the spill as possible and reuse it as it was intended. It may be necessary to remove contaminated soil and apply it to fields.
- Have an emergency response plan for the site. Know where the runoff water will flow, how to handle your particular fertilizers and whom to call for help.

Cleanup and Disposal Practices

Recommendations

- Minimize container and equipment cleanup needs by mixing only enough product for the job at hand.
- Always start with a clean, calibrated sprayer that can provide appropriate agitation.
- Clean containers and equipment at the application site if you don't have a mixing and loading pad. Put the rinsate in your spray tank and apply evenly over a field or dispose of as the label specifies.
- Rinse product containers immediately. Delays can result in residue drying and becoming much more difficult to remove.
- Stay, at least 100 feet, away from all wells including abandoned wells, drainage wells and sinkholes when mixing, loading, rinsing or washing containers and equipment unless these operations are carried out on an impervious pad.

References: Michigan University Extension, Farm.A.Syst, Improving Fertilizer Storage and Handling

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Emergency Response Plan

Suggested Format for an Emergency Response Plan

☐ **Emergency response contact list**
- ➞ Facility personnel
- ➞ Emergency assistance numbers

☐ **Product Labels**
- ➞ A complete copy for each pesticide and fertilizer stored at the facility

☐ **Product material safety data sheets (MSDS)**
- ➞ A complete copy for each pesticide and fertilizer stored at the facility

☐ **First aid Information**

☐ **Pre-fire planning**
- ➞ Invite local fire department to inspect facility annually. Familiarize them with the facility and its storage area. Provide them with the names and numbers of persons to be contacted in case of a fire.

☐ **Maps**
- ➞ Maps of the facility that include:
  - ⇨Buildings;
  - ⇨Pesticide/Fertilizer storage areas;
  - ⇨Mixing, loading and rinsate recycling areas;
  - ⇨Vehicle parking
  - ⇨Sanitary sewer inlets, storm water inlets and outlets and wells
- ➞ Facility map key and scale
- ➞ Map of surrounding area
- ➞ Surrounding area map key

☐ **Use and handling procedures**
- ➞ Procedures should thoroughly detail the facility's pesticide/fertilizer handling practices and rinsate use, including container rinsing and disposal methods and equipment (i.e. type of backflow prevention device being used.)

☐ **Emergency equipment and supplies for pesticide and fertilizer incidents**
- ➞ Identify available, working personal protective equipment and supplies. Specify location(s) at facility where the materials are stores.
- ➞ Identify available, working emergency equipment and supplies. Specify location(s) at the facility where these materials are stored.
- ➞ List of emergency contractors.

☐ **Release Procedures**
- ➞ Thoroughly describe the facility's pesticides and/or fertilizer release response procedures and practices, including use and/or disposal of spilled materials.

☐ **Employee release response training**
- ➞ Document employee name and date they completed training

☐ **Date last revised/updated**
- ➞ The emergency response plan should be kept current. It should reflect any changes in storage, handling or disposal practices and procedures.

References: http://www.mda.state.mn.us/incidentresponse/responseplan.htm;

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Emergency Response Plan
- Contact List

EMERGENCY PHONE LIST

Farm Owner: _____ Contact Number: _____

Farm Operator: _____ Contact Number: _____

Farm Manager: _____ Contact Number: _____

Other:

Name, affiliation: _____ Contact Number: _____

Who Do I Call?

Fire Department: _____	**Phone Number**: _____
EMS/Ambulance: _____	**Phone Number**: _____
HAZ-MAT: _____	**Phone Number**: _____
Police (Local): _____	**Phone Number**: _____
Police (State): _____	**Phone Number**: _____
Hospital: _____	**Phone Number**: _____
Municipality: _____	**Phone Number**: _____
Electric Utilities: _____	**Phone Number**: _____
Gas Utilities: _____	**Phone Number**: _____
Cable Utilities: _____	**Phone Number**: _____
Pesticide Spills: **Chem-Trec**	**Phone Number:** 1 (800) 424 - 9300
Poison Control Center	**Phone Number**: 1(800) 4poison 1(800) 476-4766
DIG SAFELY	**Phone Number**: 1 (800) 272 - 4480
NYS DEC Emergency Spill Hotline (24hour)	**Phone Number**: 1 (800) 457 - 7362

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This list was formulated by Suffolk County Soil & Water Conservation District

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Emergency Response Plan
- Plan of Action for:
Pesticide Spills
Petroleum Spills

Farm Emergency Plan

Name: _____

Address: _____

Telephone: _____

To prepare for an event such as a petroleum spill, pesticide spill or any other occurrence that conveys contaminants into surface or groundwater, I will make the following equipment available to contain and control the spill:

_____

_____

_____

The steps that I will take if any spill occurs are:

Eliminate the source of the spill (if possible):

_____

_____

Contact the **NYS DEC 24 Hour Spills Hotline 1(800) 457-7362.**

Contact the local municipality: _____

Contact the Fire Dept. (If risk of fire or explosion): _____

Contain contaminated runoff and prevent water quality impairment with the following methods:

_____

_____

Clean up the spill, decontaminate if needed, and remove contaminates based on advice from the NYS DEC or the Material Safety Data Sheets (MSDS's).

References: Suffolk County Soil and Water Conservation District Farm Emergency Planning Packet

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